

Rohit Kumar

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PROFESSIONAL SUMMARY

MCA graduate and Python/AI Developer with hands-on expertise in Generative AI (RAG, LangChain, Local LLMs), Computer Vision (YOLOv8, OpenCV), and full-stack integration (FastAPI, Django, React.js). Proven track record of building and deploying production-grade AI pipelines, persistent vector stores, and Dockerized microservices.

TECHNICAL SKILLS

- Languages: Python, JavaScript, SQL
- Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, ChromaDB (Vector DB), Ollama (Llama 3), Dense Embeddings (nomic-embed-text), Prompt Engineering
- AI & Machine Learning: PyTorch, TensorFlow, OpenCV, YOLOv8, CNNs, Tesseract OCR, Scikit-Learn
- Frameworks & Web: FastAPI, Django, React.js, Node.js, RESTful APIs, Streamlit
- DevOps & Tools: Docker, Docker Compose, Git/GitHub, Postman, Linux, MySQL

WORK EXPERIENCE

Python AI Intern | Excellence Technology

Jan 2025 – June 2025

- Utilized Python and the Django framework to build and maintain scalable, robust web applications.
- Leveraged Deep Learning, YOLOv8, and OpenCV to develop AI-driven solutions, including real-time image recognition and automated detection systems.
- Designed interactive dashboards and generated data insights using Power BI and Tableau to facilitate data-driven decision-making.
- Successfully designed, developed, and deployed two major projects—SmartCropCare (Crop Disease Detection) and Smart Parking System—from concept to final deployment.

KEY PROJECTS

Enterprise Offline RAG & Document Intelligence System | [GitHub](#)

Technologies: Python, FastAPI, LangChain, ChromaDB, Ollama (Llama 3), Streamlit, Docker Compose

- Built a 100% offline, privacy-compliant RAG system with zero data leakage using local Ollama (Llama 3) and LangChain.
- Engineered PDF ingestion pipeline chunking dense texts and storing 768-dim embeddings in persistent ChromaDB.
- Implemented multi-turn conversational memory with contextual history and deduplicated citation source attribution.
- Containerized frontend (Streamlit) and backend (FastAPI) as isolated microservices orchestrated via Docker Compose.

Crop disease detection and prevention using AI | [GitHub](#)

Technologies: Python, Django, MySQL, HTML/CSS

- Trained a deep CNN model achieving 94%+ validation accuracy across 20+ plant disease classes using TensorFlow.
- Engineered a geo-targeted advisory module providing location-based seasonal prevention tips and treatment strategies.

Smart Parking System Using AI | [GitHub](#)

Technologies: Python, Django, YOLOv8, OpenCV, Tesseract OCR

- Architected an Automatic Number Plate Recognition (ANPR) pipeline via YOLOv8 and Tesseract OCR, slashing check-in latency by ~60%.
- Integrated RBAC employee authentication, manual gate override safeguards, and automated dynamic PDF receipt generation.

EDUCATION

Master of Computer Applications (MCA)

GNA University, Phagwara | Year of Passing: 2025 | CGPA: 8.12

Bachelor of Science Information Technology

GNA University, Phagwara | Year of Passing: 2023 | CGPA: 7.22